



Nanotech in medicine

Carbon Nanotubes are now used with vaccines and drugs for gene therapy and tissue regeneration. <http://digit.in/mar21-23>



Less vehicle carbon emissions

Breakthroughs in the steel industry now make it possible to lower CO2 emissions from cars. <http://digit.in/mar21-24>

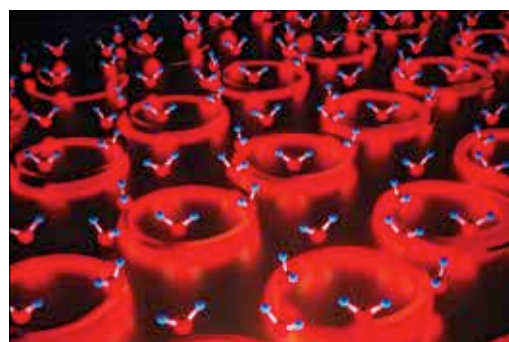
do this, they use a technique called x-ray scattering, where the Complex Materials Scattering (CMS) beamline at BNL's facility is used to fire a high intensity x-ray beam at low incidence towards the surface of the sample. The x-ray beam reflects off the surface in a characteristic pattern, providing snapshots of the structures within the blend along 5 millimeter long strips. The beamline is used to characterise a wide range of materials including biomaterials, composite materials, soft matter and nanomaterials. Together, the electrospray deposition tool and the beamline form a platform that allows research to rapidly prototype new material blends, and characterise them. The team of researchers tested the platform by using three types of self-assembling polymers, but the same platform can also be used on polymers, nanoparticles and small molecules. Characterisation of the sample showed that there were highly organised morphologies that emerged at different blend compositions. At various points in the sample, the polymers assembled into structures that are described as sheets, cylinders and spheres. Researchers now hope to create the second generation of the platform, building an electrospray device that can blend more than three components, and capture the morphological structures forming at the

time of the deposition process itself. The finding promises to accelerate research into how colloids, granular matter, liquid crystals, polymers, and active matter can all be blended together for future applications.

LIGHT BLENDED WITH MATTER

Infrared probes are very useful for characterising materials, because how materials absorb infrared light depends on the atomic composition of the material, and the infrared absorption of the material acts as a fingerprint for its chemical composition. The sensors that probe this composition can be improved by increasing the strength of the interaction of the infrared light with the atomic vibrations of the materials. This can be accomplished by trapping the light in small volumes that contain the materials, which can be accomplished by simply using a pair of mirrors to bounce the light back and forth, but much more effectively by using nanometer scale metallic structures called nanocavities.

Researchers from the University of Minnesota developed a new process for producing a quantum state that is partially light and partially matter, which can provide insights into developing the next generation of optical and electrical devices that use quantum phenomena. The researchers achieved ultrastrong coupling between infrared light and atomic vibrations, or matter. The light was trapped in tiny 2 nanometer wide annular holes in a thin wafer of gold. These cavities are approximately 25,000 times smaller than the width of a single human hair. Filling the cavities were silicon dioxide, which is the same as the glass used in a window. Essentially, each of these



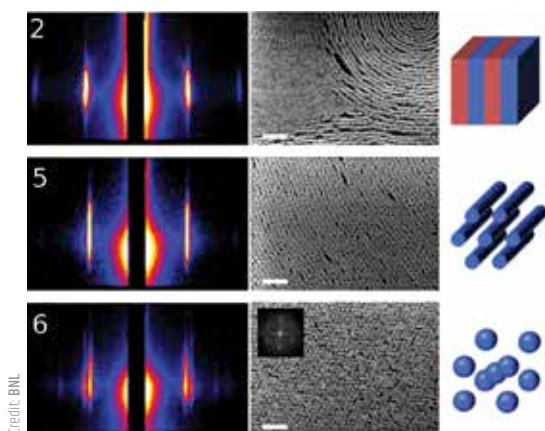
Credit: Oh Group, University of Minnesota

Annular holes in a thin gold film that enable ultrastrong coupling between light and matter

tiny cavities is a nanoscale version of the coaxial cables that provide electricity to electronic devices. The material was created using modified fabrication techniques used in the semiconductor industry.

Sang-Hyun Oh, senior author of the study says, "Others have studied strong coupling of light and matter, but with this new process to engineer nanometer-sized version of coaxial cables, we are pushing the frontiers of ultrastrong coupling, which means we are discovering new quantum states where matter and light can have very different properties and unusual things start to happen. This ultrastrong coupling of light and atomic vibrations opens up all kinds of possibilities for developing new quantum-based devices or modifying chemical reactions."

Within the cavities something remarkable takes place. The absorbed energy is rapidly transferred back and forth between the photons in the nanocavities and the phonons, or atomic vibrations in the materials, that the light photon and the matter phonon cannot be distinguished, and merge into new entities that are simultaneously light and vibrations, known as "polaritons". In the coupling regime, the vacuum is not entirely empty, and the stronger the coupling is, the stranger the quantum mechanical effects observed. The breakthrough may one day allow researchers to create light from vacuum, and find practical applications for the so called zero energy fluctuation in vacuum. 



Credit: BNL

X-ray scattering patterns on the left, scanning electron microscopy images in the middle, and morphology schematics on the right, showing the formation of vertical sheets, cylinders and spheres. The white bar for scale is 200 nanometers.